

Medical Robotics

Prof. Marilena Vendittelli
Sapienza Università di Roma

Lecture 6

Less Conventional Kinematic Architectures

Examples

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1. Recap and Lecture Overview

This lecture completes the illustration of less conventional architectures with two detailed design examples in which parallel mechanisms are used not merely because of their well-known accuracy advantages but to optimally transmit the motion of piezoelectric actuators. Piezoelectric stack actuators offer very high mechanical bandwidth (hundreds of Hz to kHz range), no backlash, and no friction in their actuation principle — making them well suited to the high-frequency, high-precision compensation required. Their main limitation is a short stroke (typically 100–200 μm), which motivates the amplification mechanisms described in this lecture. In practice, their displacement–voltage characteristic exhibits hysteresis of roughly 10–15%, which must be compensated in precision control applications. In the first example, a parallel architecture is used in a subtle and deliberately counter-intuitive way: *proximity to a parallel singularity is exploited to amplify the stroke of piezoelectric actuators*.

A recurring theme. Both design examples in this lecture (Cardiolock and the Pentagon mechanism for *in vivo* microscopy) were research prototypes. The distinction between a clever idea and a deployable medical device is a recurring lesson of this course: good technical ideas must ultimately satisfy safety, regulatory, and cost–benefit constraints that research prototypes often do not meet.

2. Example 1: The Cardiolock Active Heart Stabilizer

2.1. Clinical Motivation

Traditional open cardiac surgery requires stopping the heart and establishing extracorporeal circulation (ECC) — a pump that oxygenates and circulates the blood while the heart is arrested. ECC carries significant morbidity: the risk of stroke, neurological impairment, and vascular damage is substantially elevated compared to beating-heart procedures.

Beating-heart surgery avoids these risks but introduces a new challenge: the surgical target is in continuous motion due to the cardiac cycle. A **passive mechanical stabilizer** (such as the Esculap FlexHeart or Medtronic Octopus devices) is placed on the heart surface to reduce local motion, allowing the surgeon to work on a small stabilized area. However, passive stabilizers leave *residual motions* of the order of a few millimetres. For many procedures, including coronary bypass anastomosis, the required precision is of the order of 0.1 mm. The residual motion of passive devices is therefore clinically unsatisfactory, motivating the development of an **active stabilizer** that compensates for the residual motion in real time.

Why technology matters beyond technical performance. One rationale for robotic assistance in surgery is not merely precision but *uniformity*: the quality of the surgical outcome becomes less dependent on the individual surgeon’s skill and experience. A technology that narrows the gap between an expert and a novice surgeon has broad healthcare implications beyond the operating theatre.

2.2. System Architecture of the Cardiolock Device

The Cardiolock device was designed for a minimally invasive context: the active stabilizer enters the body through a trocar (a small port on the chest wall), satisfying the RCM constraint. The system consists of:

- A **positioning system** (external arm) to coarsely place the device.
- A **stabilizer shaft**: a long, rigid rod inserted through the trocar.
- **Stabilizer fingers** at the distal tip, which rest on the heart surface.

- A **high-speed camera with endoscope** for visual feedback.
- An **active structure** at the base of the shaft providing two compensating degrees of freedom.

The compensation principle is straightforward: when the heart moves in a given direction due to the cardiac cycle, the active structure moves the base of the shaft in the *opposite* direction. By lever action along the long shaft, this motion at the base is transferred to a compensating motion at the stabilizer fingers. Finite element analysis showed that, with a shaft length of $L = 250$ mm, the residual deflection at the tip due to shaft flexibility can be reduced to the sub-millimetric range.

2.3. Kinematic Architecture: Two-DOF Serial Spherical Mechanism

The compensation requires two end-effector degrees of freedom in directions perpendicular to the stabilizer shaft (i.e., motions in the x - and y -directions at the tip). Since the required tip displacements are small (1–2 mm) and the shaft is long (250 mm), these displacements can be well approximated by two small-angle rotations about the RCM point.

Two revolute joints J_1 and J_2 are mounted at the shaft base with their axes forming angles α_1 and α_2 with respect to the shaft axis, in two orthogonal planes. This geometric arrangement decouples the two compensation directions: rotation about J_1 produces motion in the x -direction only, and rotation about J_2 produces motion in the y -direction only (to first order). The differential kinematics relating joint velocities to end-effector velocities is:

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} L \sin(\alpha_1) & 0 \\ 0 & L \sin(\alpha_2) \end{pmatrix} \begin{pmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{pmatrix}$$

The angles α_1 and α_2 are design parameters: larger angles increase the tip displacement per unit of joint rotation (better compensation range) but also increase the inertia of the moving structure (worse dynamic performance). A trade-off is required.

2.4. Actuation Problem and the Role of Parallel Mechanisms

The compensation must occur at the frequency of the cardiac cycle (approximately 1 Hz, but with higher harmonics), requiring actuators with *low inertia, no backlash, and high mechanical bandwidth*. Classical electric motors, with their gear trains and associated backlash, do not satisfy these requirements.

Piezoelectric stack actuators offer the required dynamic properties: they have no backlash, no friction, and sub-millisecond response times. Their limitation is that they deliver only *linear motion* with a *limited stroke* (of the order of 100–200 μm for typical stacks). Two problems must therefore be solved:

1. Convert linear (translational) motion into rotational motion for joints J_1 and J_2 .
2. Amplify the insufficient stroke of the piezo actuators.

A simple slider-crank mechanism can convert translation to rotation, but cannot provide a high rotation-to-translation ratio without sacrificing stiffness in certain directions. The solution adopted in Cardiolock is a **planar parallel structure operating near a type-2 (parallel) singularity**.

2.5. The 3-PRR Parallel Structure as a Stroke Amplifier

The key insight is the following. Recall that at a type-2 (parallel) singularity, the end-effector can undergo infinitesimal motion even with all actuators locked. In the vicinity of such a singularity,

the ratio between the end-effector velocity and the actuator velocity can be made arbitrarily large. In other words, *proximity to a parallel singularity amplifies actuator motion*.

Each of the two joints J_1 and J_2 is realized by a 3-PRR planar parallel structure (three legs, each consisting of one *actuated prismatic* joint followed by two passive revolute joints). In the chosen configuration, legs 1 and 3 are blocked (their actuators set to zero velocity), while actuator u_2 controls the rotation of the end-effector platform.

The singular configuration of this 3-PRR is reached when all legs are aligned with the bisectors of the triangular end-effector. Close to this singularity, if $u_1 = u_3 = 0$, the end-effector translation $(\dot{x}, \dot{y})$ is zero but a pure rotation $\dot{\theta}$ is possible. The relationship between the actuator velocity $\dot{q}_2$ and the end-effector rotational velocity $\dot{\theta}$ is:

$$\dot{\theta} = \frac{1}{\|EB_2\| \sin(\epsilon)} \dot{q}_2$$

where ϵ is the angle between the second leg and the line EB_2 (with E the centroid of the end-effector triangle and B_2 the platform attachment point of leg 2). The smaller ϵ is, the closer the structure is to the singularity, and the **larger is the amplification** of $\dot{q}_2$ into $\dot{\theta}$. The parameter ϵ is a *design parameter* that controls the velocity gain.

Summary of the Cardiolock actuation chain.

Piezo actuator (linear motion) $\xrightarrow{\text{3-PRR near singularity}}$ Amplified rotation of J_i $\xrightarrow{\text{serial spherical kinematics}}$ Tip motion compensation

Each revolute joint J_i of the serial spherical architecture is realized by one 3-PRR parallel structure. The amplification is tunable through the geometric parameter ϵ .

2.6. The Jacobian of the 3-PRR: Derivation via the Equiprojective Property

The pose of the end-effector $\mathbf{X} = (x, y, \theta)^T$ is determined by the orientation θ and the position (x, y) of the centroid E .

The Jacobians $\mathbf{J}_q$ and $\mathbf{J}_X$ in the velocity equation $\mathbf{J}_q \dot{\mathbf{q}} = \mathbf{J}_X \dot{\mathbf{X}}$ are derived using the **equiprojective property**: for any two points A_i and B_i on the same rigid body, the projections of their velocities onto the line $A_i B_i$ must be equal:

$$\mathbf{v}_{A_i} \cdot \mathbf{A_i B_i} = \mathbf{v}_{B_i} \cdot \mathbf{A_i B_i}$$

For the 3-PRR, $\mathbf{v}_{A_i} = \dot{q}_i \mathbf{u}_i$ (the actuator velocity along its prismatic direction) and $\mathbf{v}_{B_i} = \dot{\mathbf{v}}_E + B_i E \times \dot{\theta}$. Applying this for $i = 1, 2, 3$ and collecting terms into matrix form:

$$\mathbf{J}_q = \begin{pmatrix} \mathbf{u}_1 \cdot \mathbf{A_1 B_1} & 0 & 0 \\ 0 & \mathbf{u}_2 \cdot \mathbf{A_2 B_2} & 0 \\ 0 & 0 & \mathbf{u}_3 \cdot \mathbf{A_3 B_3} \end{pmatrix} \quad \mathbf{J}_X = \begin{pmatrix} A_1 B_1|_x & A_1 B_1|_y & \mathbf{A_1 B_1} \times B_1 E|_\theta \\ A_2 B_2|_x & A_2 B_2|_y & \mathbf{A_2 B_2} \times B_2 E|_\theta \\ A_3 B_3|_x & A_3 B_3|_y & \mathbf{A_3 B_3} \times B_3 E|_\theta \end{pmatrix}$$

2.7. Cardiolock 2: Compliant Mechanisms

In the Cardiolock 2 prototype, the 3-PRR parallel structures were implemented as **compliant mechanisms** — flexible structures without discrete joints, where deformation of elastic elements replaces rigid-link rotation. The **Pseudo-Rigid Body Model (PRBM)** approach allows the analytical kinematic framework developed for rigid-link parallel mechanisms to be applied directly to these compliant designs, by replacing each flexure with an equivalent revolute joint and a torsional spring.

2.8. Why Cardiolock Was Not Adopted Clinically

Despite achieving sub-millimetric compensation in animal trials, the Cardiolock system was not adopted in clinical practice. The primary barriers were:

- **Safety concerns:** the active structure moves rapidly (at cardiac frequency, roughly 1 Hz with significant harmonics) inside the patient's chest. Any control failure could result in high-velocity uncontrolled motion near the heart.
- **Bulk and complexity:** the parallel mechanism at the base of the stabilizer shaft was mechanically complex and relatively bulky for an intraoperative device.
- **Cost–benefit balance:** the marginal improvement over high-quality passive stabilizers, combined with the regulatory burden and clinical risk of a novel active device, did not justify widespread adoption.

3. Example 2: Image Stabilization for *In Vivo* Microscopy

3.1. Clinical and Scientific Motivation

Molecular imaging (and in particular **confocal fluorescence microscopy**) allows the visualization of biological processes at the cellular and sub-cellular level within living organisms. Spatial resolutions in the sub-micrometre range are achievable. At this scale, physiological motion — heartbeats (amplitude: 20–40 μm at frequencies of ~ 10 Hz) and breathing (amplitude: 1000–1200 μm at ~ 0.2 Hz) — causes the region of interest to move out of the focal plane, severely degrading or entirely destroying image quality.

The solution is a fast, active mechanism that synchronizes the position of the objective lens with the motion of the tissue, keeping the focal plane locked onto the target at all times.

3.2. The Pentagon Mechanism

The chosen actuator is again a **piezoelectric stack** (Physik Instrumente P239.90, stroke 180 μm), for the same reasons as in the Cardiolock case: no backlash, no friction, high bandwidth. The stroke, however, is insufficient: the required compensation range is of the order of 500 μm (to handle breathing motion).

The amplification mechanism is a **five-bar closed-loop linkage** (the “Pentagon”), driven by two piezo actuators pushing on the first two links (l_1 and l_2 at the base). The objective lens is mounted at the apex of the Pentagon (position $\mathbf{r}$). The two actuated joints are θ_1 (pushed by actuator 1 via l_1) and θ_2 (pushed by actuator 2 via l_2). The remaining three joints (θ_3 , θ_4 , θ_5) are passive. In the physical implementation, the links are **flexible segments** (compliant mechanism), with the corners of the Pentagon acting as flexure joints.

Pentagon vs. Cardiolock: a common principle. Both examples use a closed kinematic chain driven by piezo actuators to amplify an insufficient stroke. The design objective in both cases is to find link geometries that yield the required amplification (the right velocity gain) while maintaining structural stiffness and dynamic performance. The mathematical tool for achieving this is the Jacobian of the mechanism.

3.3. Jacobian Computation via the Virtual Tree Method

To compute the Jacobian $\mathbf{J}_{\text{rob}}$ relating the lens velocity $\dot{\mathbf{r}}$ to the active joint velocities $\dot{\theta}_G = (\dot{\theta}_1, \dot{\theta}_2)^T$, the closed chain is opened by *virtually cutting* one of the links at the apex where the lens is placed. This transforms the closed chain into two open-chain branches:

- Branch 1: joints θ_1 (active), θ_3 (passive).
- Branch 2: joints θ_2 (active), θ_4, θ_5 (passive).

The velocity of the lens can be expressed from either branch:

$$\dot{\mathbf{r}} = \begin{pmatrix} J_1 & J_3 \end{pmatrix} \begin{pmatrix} \dot{\theta}_1 \\ \dot{\theta}_3 \end{pmatrix} = \begin{pmatrix} J_2 & J_4 & J_5 \end{pmatrix} \begin{pmatrix} \dot{\theta}_2 \\ \dot{\theta}_4 \\ \dot{\theta}_5 \end{pmatrix}$$

where $J_i = \partial \mathbf{r} / \partial \theta_i$. Equating the two expressions and grouping the passive joints $\boldsymbol{\theta}_S = (\theta_3, \theta_4, \theta_5)^T$ and the active joints $\boldsymbol{\theta}_G = (\theta_1, \theta_2)^T$:

$$J_S \dot{\boldsymbol{\theta}}_S = -J_G \dot{\boldsymbol{\theta}}_G$$

with $J_S = (J_3 \quad J_4 \quad J_5)$ and $J_G = (J_1 \quad J_2)$. The passive joint velocities can be expressed as:

$$\dot{\boldsymbol{\theta}}_S = -J_S^\# J_G \dot{\boldsymbol{\theta}}_G$$

Substituting back into the velocity expression for Branch 1:

$$\dot{\mathbf{r}} = \begin{pmatrix} J_1 & J_3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ J_A & \end{pmatrix} \dot{\boldsymbol{\theta}}_G = \mathbf{J}_{\text{rob}} \dot{\boldsymbol{\theta}}_G$$

where J_A is the first row of $-J_S^\# J_G$.

3.4. Design Optimization via the Jacobian

The design requirements for the Pentagon are:

1. **Sufficient amplification:** the ratio between lens velocity and actuator velocity must be large enough to cover the required compensation range.
2. **Isotropy:** the amplification should be as uniform as possible in all directions of the lens plane, so that the mechanism does not favour one compensation direction over another.

Both properties are controlled through the eigenvalues of $\mathbf{J}_{\text{rob}}$: the amplification ratio is related to the eigenvalues, and isotropy is achieved by maximizing the minimum singular value (minimizing the condition number) of $\mathbf{J}_{\text{rob}}$.

The design parameters are the link lengths $l_1, \dots, l_5$. A numerical optimization procedure was performed to find the link lengths that simultaneously provide sufficient amplification and good isotropy, subject to the geometric constraint that the total linkage size must fit within the microscope setup.

Result: the Pentagon amplifies the stroke of the piezo actuators from $180 \mu\text{m}$ to $563 \mu\text{m}$ (actuator 1) and $464 \mu\text{m}$ (actuator 2). These values are sufficient to compensate both heartbeat-induced motion and, with some margin, breathing-induced motion.

4. Summary and Key Conceptual Threads

- **Parallel singularities can be exploited, not only avoided.** Near a type-2 (parallel) singularity, the ratio of end-effector velocity to actuator velocity diverges.
- **The design parameter ϵ (Cardiolock) and the link geometry (Pentagon) tune the amplification gain.** In both cases, the Jacobian of the mechanism is the central analytical tool for the design.

- **Compliant mechanisms and the PRBM.** When discrete revolute joints are replaced by elastic flexures (compliant mechanisms), the kinematic analysis developed for rigid-link parallel mechanisms can still be applied via the Pseudo-Rigid Body Model.

Key topics for revision.

1. The three types of singularity in parallel manipulators and their consequences for medical robot design (particularly the danger of type-2 singularities inside the workspace).
2. The Cardiolock system: clinical motivation, kinematic architecture, the role of the 3-PRR parallel structure, and why proximity to a parallel singularity amplifies the piezo actuator stroke.
3. The Pentagon mechanism: clinical motivation, kinematic structure, the Jacobian derivation via the virtual tree method, and how the link geometry is optimized for sufficient and isotropic amplification.
4. Why both Cardiolock and Pentagon use piezo actuators and what their limitation (stroke) motivates.

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